

SUSHANT KUMAR

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PROFESSIONAL SUMMARY

Biomedical Engineering graduate and Projects Engineer at IRDE, DRDO, working on AI-based computer vision and electro-optical systems for thermal imaging. Trained and benchmarked deep learning models on thermal datasets with Python and ONNX/C++ inference on CPU and GPU, optimized for NVIDIA Jetson. Experienced in real-time thermal image processing, multisensor fusion, calibration and testing, with a background in medical equipment QA and instrumentation.

TECHNICAL SKILLS

Programming & Tools: Python, C, C++, MATLAB, CUDA; Autodesk 360, 3D Printing

Deep Learning Models: YOLOv11, YOLO26, SSD, EfficientNet, EfficientNet-Lite, InternImage, ViT, VGG

Frameworks & Deployment: TensorFlow, OpenCV, ONNX, TensorRT, NumPy, Pandas

Thermal Imaging & Embedded: MWIR thermal imaging, NVIDIA Jetson (AGX Orin, Nano), turbulence correction, stabilization, multisensor fusion, NUC/BPR, calibration

Medical Devices & Domain: ECG, Defibrillator, ICU Ventilator, Multiparameter Monitor, MRI, Signal Processing, Sensors, QA & Testing

PROFESSIONAL EXPERIENCE

Projects Engineer | IRDE-DRDO, Dehradun

Sep 2026 – Present

- Developed CUDA-based atmospheric turbulence correction on Jetson AGX Orin for real-time performance and implemented real-time DNN-based digital image stabilization for thermal imaging systems.
- Implemented NUC and BPR thermal image enhancement and developed thermal panorama generation for wide-area surveillance.
- Performed multisensor alignment and fusion of thermal and CCD cameras; integrated electro-optical payloads and conducted system-level calibration, validation and testing.

AI/ML ENGINEER | COMPUTER VISION | QA Engineer | IRDE-DRDO, Dehradun

Nov 2025 – Sep 2026

- Worked on AI-based computer vision and automated detection projects.
- Developed and annotated multi-environment thermal object-detection datasets; trained YOLO26 models on NVIDIA A100 and optimized them with TensorRT on Jetson AGX Orin for real-time inference.
- Trained EfficientNet, EfficientNet-Lite, InternImage and YOLO26 on thermal datasets Python inference, converted models to ONNX for C++ inference and benchmarked CPU and GPU performance.
- Supported QA activities: data validation, pipeline testing, output verification and documentation.
- Improved model reliability and performance through systematic testing and cross-functional collaboration.

AI/ML Research Intern | IRDE-DRDO, Dehradun

Jan 2025 – Feb 2025

- Developed real-time cancer cell detection pipeline on Jetson Nano using IR thermal camera and Python.
- Applied Laplacian edge detection, CLAHE, and Gaussian Blur for noise reduction and contrast enhancement.
- Optimized image processing enabling clear tumor boundary detection in IR and MRI scans.
- Contributed to algorithm design and parameter tuning, improving detection accuracy.

Medical Equipment Intern | Instrument Care Services, Kolkata

Jan 2023 – Feb 2023

- Studied working principles, block diagrams, and electronic signal flow of ECG machines, defibrillators, ICU ventilators, and dialysis units; analysed biosensor and transducer integration in life-critical devices.
- Performed hands-on fault diagnosis, calibration, and preventive maintenance on multiparameter monitors, pacemakers, and syringe pumps using oscilloscopes and electronic test equipment.
- Gained insight into AI-assisted patient monitoring trends and sensor-based early-warning systems used in modern ICU environments.
- Observed signal-conditioning circuits for SpO2, ECG, and real-time vitals tracking — bridging electronic instrumentation with AI-driven biomedical applications.

PROJECTS

Real-Time Cancer Cell Detection System | Python, OpenCV, Jetson Nano, CLAHE, Laplacian Edge Detection

- Built IR thermal + MRI based pipeline for tumor boundary detection using image enhancement techniques on Jetson Nano.

Bio-Sensor for BUN & Creatinine Detection | Quantum Dots, Fluorescence Spectroscopy, Point-of-Care

- Synthesizing and conjugating quantum dots with biomolecules for fluorescence detection of BUN and creatinine.
- Designing and fabricating a portable point-of-care device, with lab validation in progress.

Anchoring Mechanism for TMVR Device | Biomedical Design, CAD, Design Thinking, IPR

- Designed anchoring mechanism for Transcatheter Mitral Valve Replacement (TMVR) studying heart function and device principles.
- Applied design thinking at Impact Week and attended IPR workshop for biomedical innovation.

EDUCATION

B.Tech in Biomedical Engineering | North Eastern Hill University, Shillong, Meghalaya

2021 – 2025 | CGPA: 6.59

Higher Secondary (PCM) | Langat Singh College, Muzaffarpur, Bihar

2020 | 61.4%

Secondary | Shree Baldev High School, Keotsa, Bihar

2018 | 53.8%

CERTIFICATIONS & ACHIEVEMENTS

- 2nd Position – Biocraft Competition, IIT Kharagpur.
- Satellite Communication & Navigation Training – NESAC, Dept. of Space, Govt. of India (2024).
- Yuva Manthan Model United Nations (YUMN) – Delegate, policy discussions & public speaking (2024).
- International Conference on Neuroscience & Neurological Disorders – SNCI, NEHU, Shillong (2023).